H2

ANDERSON JUNIOR COLLEGE
HIGHER 2
ANSWERS

2018 PRELIMINARY EXAMINATIONS

CANDIDATE NAME	
PDG	
INDEX NUMBER	

H2 BIOLOGY 9744/03

PAPER 3

13 SEPT 2018
THURSDAY

2 hours

Long Structured
and Free-Response Questions

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graph
Do not use paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer any **one** question in the spaces provided on the Question Paper

For Examiner's Use	
1	/ 26
2	/13
3	/11
4/5	/25
	/75

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Section A

Answer **all** the questions in this section

1 Apoptosis is an important part of tissue homeostasis. It ensures that cells die quickly and are safely removed.

Every cell undergoing apoptosis shows a predictable series of changes. The cell shrinks and breaks up into dead cell fragments bounded by membranes. The membranes of these dead cell fragments display unique surface ligands, which identify the fragments to neighbouring cells ingesting these fragments.

Before the apoptotic cell breaks up, cellular DNA is degraded by endonucleases into small pieces that each differ in size by 180 base pairs.

Fig. 1.1 is an electron micrograph of a cell with ingested dead cell fragments.

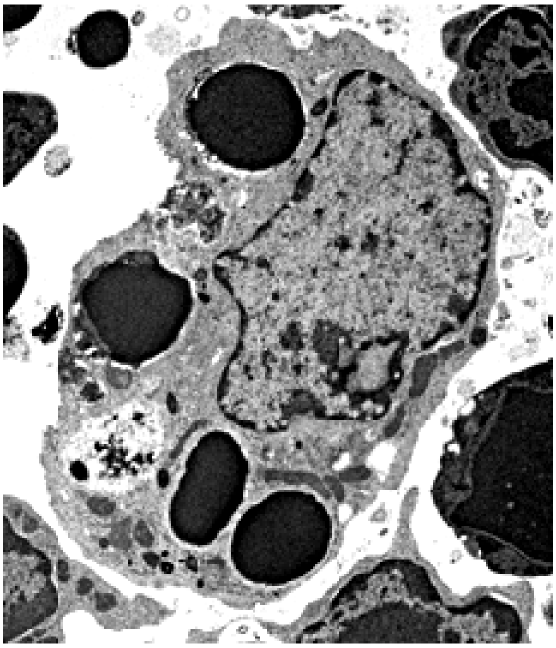


Fig. 1.1

(a) (i) Identify **one** type of cell that can ingest dead cell fragments and explain how the cell recognises these cell fragments.

- Name: macrophage, neutrophils;
- Idea that **proteins** on the outside of surface membrane of fragments **expressed** during apoptosis;
- These proteins are **complementary in shape** to the **surface receptors** on the macrophages. [3]

(ii) The DNA isolated from apoptotic cells can be analysed to confirm that apoptosis had indeed been activated.

Using the information above, outline a procedure that can be performed on a tissue extract to determine whether apoptosis has occurred.

Any 3

- Load DNA samples into the **wells of the agarose gel**; (submerged in a buffer solution).
- **DNA ladder** is loaded into a separate well to add as a **size reference**;
- **Direct current** is applied; [3]

- DNA would migrate from the **negative end** to the **positive end**;
- Stain the gel using methylene blue to visualise the band;

Compulsory point
DNA bands (reject: fragments) that differs by 180bp would be observed

(b) The understanding of apoptosis in human originated in the nematode worm *Caenorhabditis elegans.* Each worm is 1mm long and has a short life cycle. As the worm is transparent, scientists can follow how every cell in it originated from the zygote.

It was observed that in the formation of an adult worm, 1090 cells were generated, of which, exactly 131 cells died at particular points during the developmental process.

(i) With reference to the information above, suggest why *C. elegans* is especially useful as an experimental organism.

- Short life cycle – grow rapidly/large samples.
- Small size – can be kept in lab, observed under microscope.
- Transparent- used to follow the development of organism/count the numbers of cells. (must paraphrase pre-amble) [2]

(ii) A technique, known as the genetic screen, is used to identify genes that control a known process.

- For the case of *C.elegans*, scientists induced random genetic mutations by irradiating the eggs.
- The resulting adult worms were then examined.
- Analysis was then carried out to identify the genes that were being mutated.

In one of the genetic screen, it was observed that a particular *C.elegans* has more cells than normal as some cells did not die at particular points of the developmental process.

Suggest an explanation for the above observation.

- Radiation has induced **mutation in a gene that control apoptosis.**
- Leading a change in the shape/conformation/tertiary structure of the proteins controlling apoptosis
- Idea of extra cells were found in the worm as these **cells cannot undergo apoptosis.** [3]

(c) Fig. 1.2 shows one way in which apoptosis can be triggered.

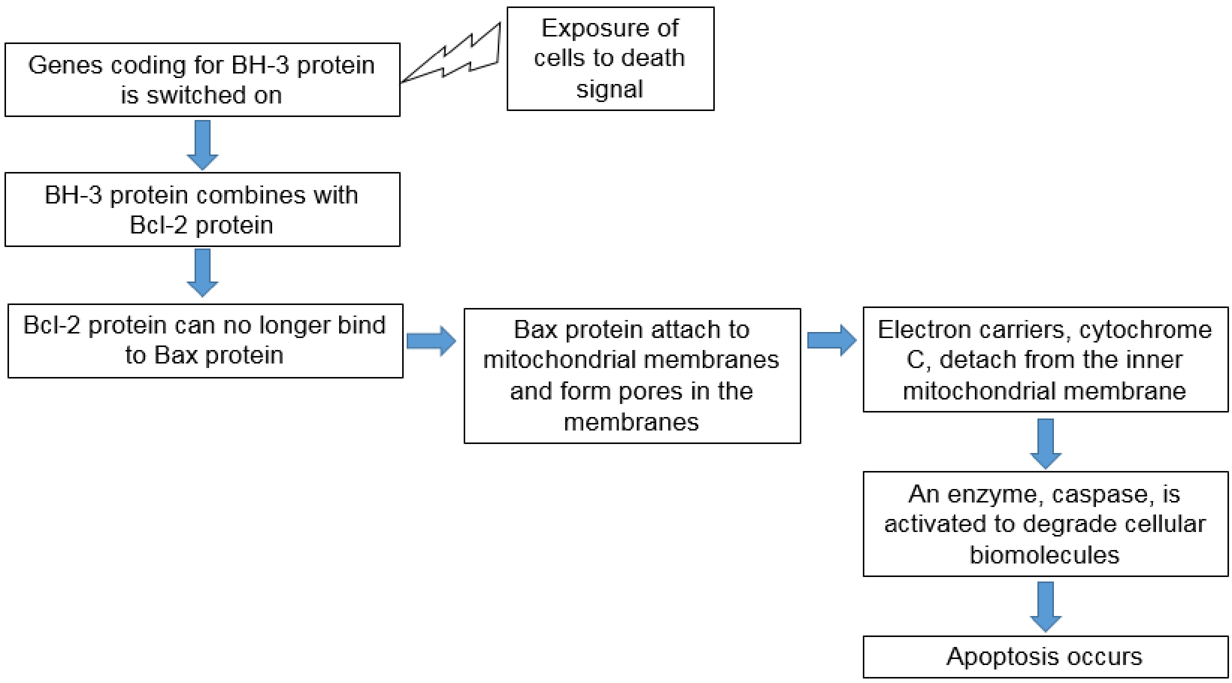


Fig.1.2

(i) With reference to Fig 1.2, predict and explain how ATP production will change when cytochrome C is detached from the inner mitochondrial membrane.

- Lower production of ATP.
- Electrons **flow through down the electron transport chain** is disrupted.
- **Proton gradient** cannot be established across the **inner mitochondrial membrane.**
- Diffusion of H+ ions from **inter-membrane space** into **matrix** through **ATP synthase** cannot occur [4]

(ii) A particular mutation leads to the excessive production of Bcl-2 protein, disrupting the apoptotic pathway, leading to B-cell lymphoma, a cancer of B-cells.

- Bcl-2 gene is found on chromosome 18.
- In B-cell lymphoma, a mutation shifts the Bcl-2 gene to chromosome 14, bringing it close to a powerful gene regulatory sequence.
- This regulatory sequence regulates the production of the immunoglobulin heavy chain.

Explain why the mutation will lead to B-cell lymphoma.

- Chromosomal translocation would place the bcl-2 gene under an **active enhancer/promoter**
- Bcl-2 gene is **transcribed** and mRNA **translated** more often.
- High concentration of Bcl-2 protein would lead to **decrease apoptosis/uncontrolled cell division**. [3]

(d) A drug, Venetoclax has been developed recently to treat B-cell lymphoma. This drug is structurally similar to a BH-3 protein.

An experiment was performed on three different patients to investigate the effect of Venetoclax on lymphoma cells.

The results of the experiment are shown in Fig. 1.3.

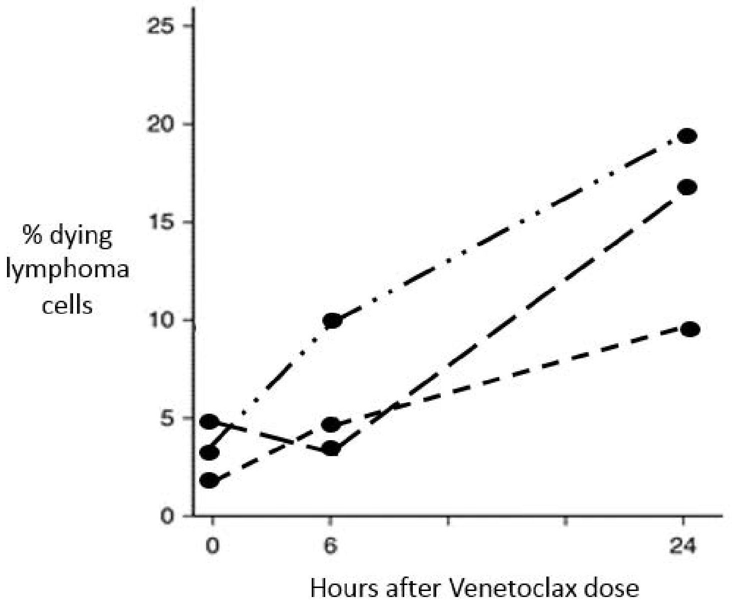


Fig 1.3

With reference to Fig. 1.3 and the information above, describe and explain the effect of Venetoclax on cancer cells.

<u>Describe:</u>

- Increase in the number of apoptotic cells + cite at least one data;
    - Data 1: from 2-3% dead cells to 8 to 9% dead cells)
    - Data 2: from 3-4% to 18 -19% dead cells
    - Data 3: from 5% to 16-17% dead cells)

<u>Explain:</u>

- Venetoclax binds to Bcl-2 protein;
- Bcl-2 no longer binds to Bax;
- Higher concentration of free Bax attach to the mitochondrial membrane/acts and release cytochrome C into the cytoplasm;
- Activates **<u>caspase</u>**, to **<u>digest cellular biomolecules</u>**, leading to **<u>apoptosis</u>** of cancer cells;

[4]

(e) Justify the idea that apoptosis is a defence mechanism in humans.

- Cytotoxic T cells induce apoptosis of virus infected cells

- To prevent **release of more mature virus** to infect other cells

- Cells with gene mutation at tumour suppressor genes/proto-oncogene are being destroyed via apoptosis.

- *Reject: apoptosis occurs to "mutated cells"/cells with DNA damage.*

- Prevent **accumulation of mutation**/**uncontrolled cell division/**
- *To accept: to prevent passing down of mutation to daughter cells*

- Telomere reached Critical length→ induced apoptosis
- to prevent genes from being eroded. [4]

[Total: 26]

2 While modern therapeutic antibody production can involve other cell models, therapeutic antibodies are conventionally produced by animals (usually mice). For many specific antibodies, animal models remain the only viable option. Fig. 2.1 shows the simplified process of antibody production in rabbits.

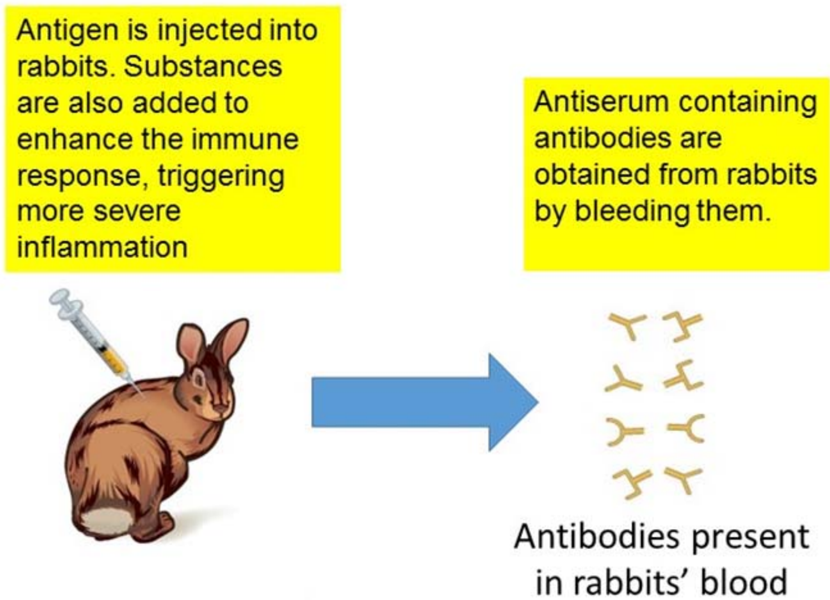


Fig. 2.1

(a) Comment on the ethical implications of using animal models for large scale antibody production.

- Method causes severe inflammation and bleeding in the animals, resulting in severe [3]

distress or discomfort in animals;
- A single animal should not be used to over-produce too many antibodies;
- Other alternatives should be explored / considered before animals are used for antibody production;

(b) Prokaryotic cells can potentially be used for antibody production. Fig. 2.2 shows a possible experimental procedure for antibody production in bacteria cells.

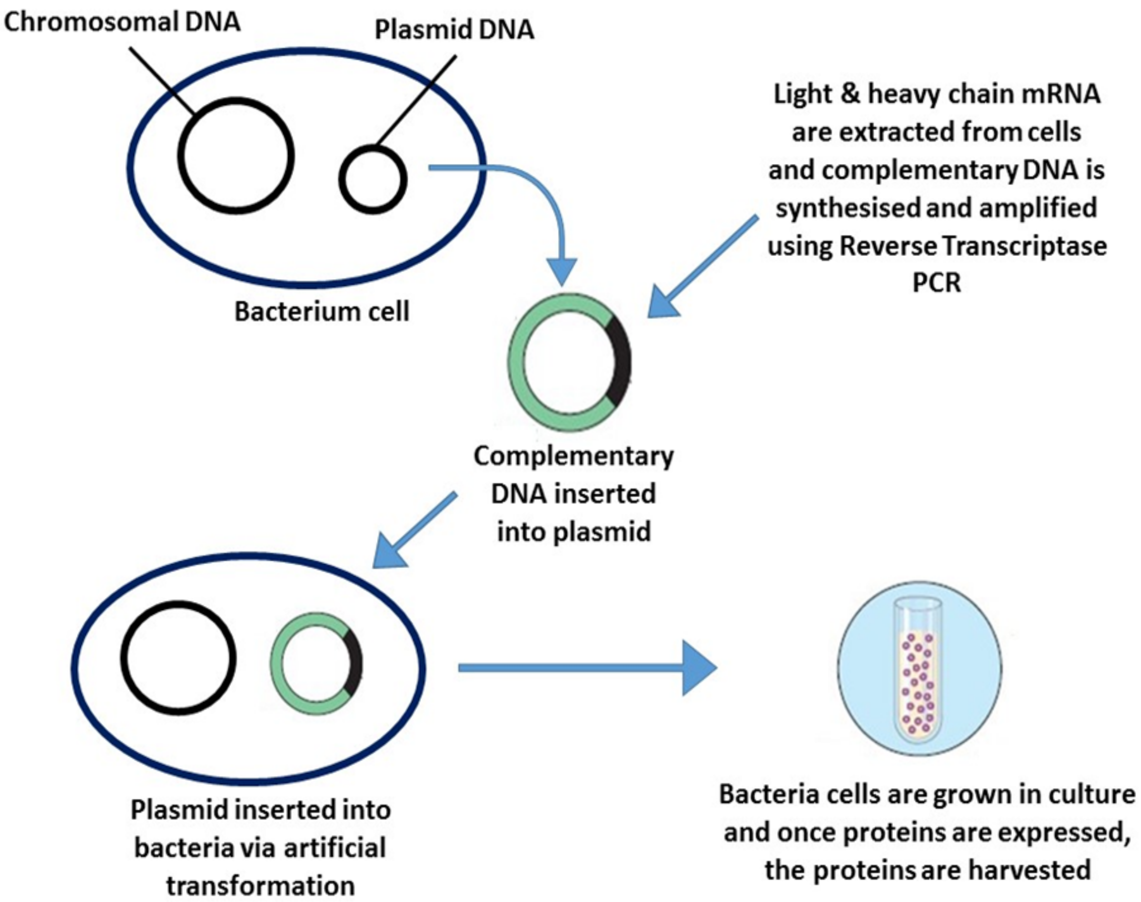


Fig. 2.2

To obtain DNA sequences of the light and heavy chain genes, light and heavy chain mRNA are extracted from plasma cells and then DNA containing the gene sequences are produced using reverse transcriptase Polymerase Chain Reaction (PCR).

The procedure for reverse transcriptase PCR is as follows:

- The extracted mRNA is mixed with DNA nucleotides, reverse transcriptase and incubated at 65°C for 5 minutes.
- RNase is then added to the reaction mixture and incubated at 42°C for 50 minutes, then 70°C for 15 minutes.
- Primers, *Taq* polymerase and DNA nucleotides are then added. The first cycle of PCR is carried out at 72°C, before the normal three-stage PCR cycle is carried

out for the next 40 cycles at temperature 95ºC, 60ºC and 72ºC.

(b) Explain the advantages of extracting mRNA instead of the DNA gene sequence from the genome of plasma cells.

- Ref. to the idea that there are more copies of mRNA and DNA gene sequence in cells;
- mRNA has introns removed;
- accurately reflects the codon sequence for the actual amino acid sequence of the polypeptide chain expressed;

[max 2m] [2]

(c) For the reverse transcriptase PCR procedure, explain why

(i) extracted mRNA is mixed with DNA nucleotides and reverse transcriptase.

- Reverse transcriptase synthesises cDNA (using the mRNA as template strand); [1]

(ii) RNase is added to the reaction mixture and eventually incubated at 70ºC for 15 minutes.

- RNase degrade mRNA template strands
- RNase is then denatured by incubation 70ºC to stop degradation reaction; [2]

(iii) the first cycle of PCR is carried out at 72ºC instead of the usual 95ºC.

- cDNA template is single-stranded;
- there is no need to heat to 95ºC to separate cDNA into single strands; [2]

(d) However, production of fully functional antibodies, which originate from eukaryotic cells, in prokaryotic cells is expected to be unsuccessful.

Explain why production of fully functional antibodies in prokaryotic cells is expected to be unsuccessful.

- Lack of RER and Golgi apparatus;
- RER: to provide a conducive microenvironment for the folding of the polypeptide chain into the correct tertiary structure;
- Golgi apparatus: for the linking of the light and heavy chains by the formation of disulphide linkages; [3]

[Total: 13]

3 To protect crops against insects, many crops such as maize and cotton are genetically modified to contain a gene (Bt) from a bacterium, *Bacillius thuringiensis*. The Bt gene codes for a toxin, which inserts into the insect gut cell membrane, paralyzing the digestive tract and killing the insects which eat the crop. When these genetically modified Bt crops first became available, it was predicted that insect pests would develop resistance to these toxins.

(a) Explain how Bt resistance may arise and spread in an insect population.

- Bt resistance arise as a result of: **random mutation**s in the gene that encode proteins in the cell surface membrane / gene that encodes for protein that can bind to Bt to prevent it from inserting into the cell membrane / AVP in the insects

Any 2
- Presence of Bt toxin creates a **selection pressure** during **natural selection**
- Insects with resistance to Bt toxin/ not killed by Bt toxins are at a **selective advantage** / will be selected for
- can **survive, reproduce** and **pass** on **alleles** that confer Bt resistance to their **offspring / next generation** [4]

(b) The extent of Bt resistance in insect pest species was surveyed in 2005 and in 2011. The level of resistance in each species was classified according to the highest percentage of resistant individuals recorded in any population anywhere in the world. Three levels of resistance were identified:

- <1%
- 1–6%
- >50%

There were no reports of populations of insect pests having between 6% and 50% of resistant individuals.

The results of the surveys are shown in Table 3.1.

Table 3.1

year	total number of insect pest species surveyed	number of insect pest species susceptible to Bt toxins	number of insect pest species with reported levels of resistance		
			<1%	1–6%	>50%
2005	9	8	0	0	1
2011	13	4	3	1	5

(i) Discuss whether the data provided can support the hypothesis that rise in Bt resistance is due to introduction of Bt crops, including limitations of the data.

2 marks on data analysis

From 2005 to 2011,

- percentage of insect pest species susceptible to Bt toxins decreased from 89% to 31%

AND

- Percentage of insect pests with high level of resistance/ >50% resistance increased from 11% to 38.5%

OR

- Number of insects pests with low level of resistance / < 1% resistance increased from 0% to 23% OR Number of insects pests with low level of resistance / 1-6% resistance increased from 0% to 7%

(reject merely quoting raw values since the total number of insect pests surveyed is different- must convert to percentage)

3 marks on limitations

- Small sample size
- Survey only done in two years
- the damage done by the insect pests surveyed ;
- the number of reports of resistance for each species ;
- the proportion of populations with the highest percentage of resistant individuals ;
- the effect on the crops concerned of pest resistance at the levels given (<1%, etc.) ;e.g. the losses in yield
- the geographical spread of the insect pest species that show resistance

[4]

(ii) Justify the prediction that both insect-caused and non-insect-caused damages to Bt crops will increase with climate change.

- Extreme weather conditions- must give example e.g. drought / water logging / flood decreased growth of crops
- Increasing temperature degrade Bt toxin → Bt toxin n longer effective against insects
- Increasing temperature will increase metabolism of insects/ shorter maturation/life cycle/ reproduce in a shorten span of time → More insects to feed on crops
- increase in reproduction rate of insects , increase the development of resistance to Bt toxin

[3]

[Total: 11]

Section B

Answer **one** question in this section

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answer must be set out in parts **(a)**, **(b),** etc., as indicated in the question.

4 **(a)** Explain how fluidity of biological membranes can be maintained and the importance of fluidity to membrane function. [10]

How fluidity is maintained:

- Definition of fluidity: phospholipids and proteins free to move within layer / transversely;
- Membrane fluidity can be regulated by regulating the level of cholesterol and phospholipids with unsaturated hydrocarbon tails in the membrane;
- Membranes freeze at a lower temperature if it has a higher proportion of phospholipids with unsaturated hydrocarbon tails and cholesterol; (Accept reverse argument)
- Unsaturated hydrocarbon tails have kinks that keep the phospholipid molecules in the membrane from packing close together at lower temperatures, enhancing membrane fluidity; (Accept reverse argument)
- Cholesterol prevents close packing of phospholipids in the membrane, preventing membrane from freezing at lower temperatures;
- Membrane prevented from being overly fluid / integrity of the membrane is maintained at higher temperature as cholesterol restricts phospholipid movement through its interactions with phospholipids;
- Stabilising the lipid bilayer through hydrophobic interactions;

[max 4m]

Importance:

- ref. to need for invagination / pinching in of the membrane during cytokinesis / cell division;
- ref. to invagination of the membrane during phagocytosis / pinocytosis / endocytosis;
- ref. to formation of vesicles for the transport / trafficking of proteins from rough ER to Golgi apparatus / other membrane-bound organelles;
- ref. to formation of vesicles from *trans* face of the Golgi apparatus during protein sorting;
- ref. to fusion of vesicles from endoplasmic reticulum / containing proteins to form cis-face of the Golgi apparatus for protein modification;
- ref. to fusion of vesicles with cell surface membrane to release secretory molecules from the cell / exocytosis;
- ref. to fusion of endocytotic vesicles with lysosomes;
- ref. to membrane fluidity allowing movement of protein molecules embedded on the cell surface membrane for dimerisation of receptors;
- ref. membrane fluidity to allow locomotion of unicellular organisms e.g. amoeba;
- ref. diffusion of small or non-polar molecules through the membrane, e.g. water / oxygen / carbon dioxide (at least one named example);
- ref. to transmembrane transport proteins changing conformation, such as sodium-potassium pump / ligand-gated sodium ion channel / voltage-gated sodium ion channel (at least one named example);

[max 4m]

Examiners' comments:
There were many answers which incorrectly described fatty acid chains instead of fatty acid tails of phospholipids. Many students also confused phospholipids with cholesterol, describing cholesterol as have kinks or being unsaturated or saturated. There were also students who describe fatty acid chains which were branched.

Many answers included general statements about membrane functions in cells instead of relating fluidity to these functions. The general statements were not credited. Also, merely mentioning that 'membrane fluidity is important for endocytosis or exocytosis' is insufficient. It is important to explain clearly what the membrane fluidity was critical to, i.e. 'membrane fluidity is important for the invagination of the membrane during endocytosis' or 'membrane fluidity is essential for the secretory vesicles to fuse with the cell membrane during exocytosis' would be clearer answers which will gain credit.

(b) A human body can synthesise more than 90,000 different types of proteins. The complete set of proteins present in the various cells of an individual is known as the proteome.

A human's genome is constant, whereas the proteome of a human is constantly changing.' Discuss this statement. [15]

(discuss for human's genome is constant) due to nature of genes being DNA

- HF1 – idea of structure of DNA (sugar-phosphate back backbone and H-bonds between bases) contribute to DNA being a stable molecule?
- HF2 - How **semi-conservative DNA replication** before cell division takes place accurately/ described (during S phase of interphase) NOT mitosis
- HF3 - ***complementary base pairing*** *(A=T, C≡G)*

- HF4 - ***DNA polymerase*** *with* ***proofreading function***
- HF5 - Specific sequences have specific functions → important that the functions are maintained / Examples introns, exons, promoters, distal control elements (enhancer/ silencer), centromeres, telomeres
- HF6 - one gene, one protein hypothesis (1mRNA?)
- HF7 – How cell cycle checkpoints ensure mutations are not passed to future generations of cells Tumour suppressor gene
- HF8 - How mitosis ensures genetic stability (or give example such as spindle fibre pull sister chromatids to opposite poles) / nuclei of daughter cells are genetically identical or receive the same number and type of chromosomes as the original parent nucleus,
- HF9 - Function/ why mitosis is important: stem cells during growth of human and renewal and repair: replace dead or damaged cells by new cells which are genetically identical to original cells so that cells (used in growth/ renewal/repair) can carry out the same function;
- HF10 – idea that telomeres acts as "disposable buffer"/ gets shortened first so that genes at the chromosomal ends remain intact/ **prevents** the **loss of vital genetic information** with each round of replication/ trigger apoptosis when critical length is reach

(discuss against human's genome is constant)

- HA1 - Exposure to radiation / chemical carcinogens causes structural damage to DNA/induce mutation + *cite an example below*
    - e.g. UV light (causes thymine dimer formation)
    - e.g. chemicals (such as nitrous acid chemically reacts with base)
    - e.g. ethidium bromide (intercalates into DNA)
- HA2 - **DNA replication is not always perfect** → Spontaneous mutation – DNA polymerase adds the wrong base, and is not being rectified.
- HA3 - (nature of damage) Such structural damage causes wrong nucleotide(s) / extra nucleotide(s) / missing nucleotide(s)/ change in chromosomal structure e.g. chromosomal translocation

- HA4 - During B cell development, somatic recombination occurs/ DNA rearrangement occurs on the light and heavy chain/ VDJ segments (must have heavu and light chain?)
- HA5 - Gametic cells: Generation of gametic variation as a result of meiosis (not all cells in the same human body have same genome)
- HA6 - Process that generate variation: crossing over, independent assortment of homologous chromosomes
- HA7 - Errors in mitosis leading errors in number of chromosomes e.g. aneuploidy/ mosaic Down syndrome

- Link1 - (link between genome and proteome) Proteins are derived from protein synthesis. Their unique amino acid sequences (coded for by the DNA sequence of their genes) determine their folding thus their tertiary, quaternary structure which enables them to perform their unique functions.

(discuss for human's proteome is constantly changing)

- PF1 - need for changing protein expression as cells grow and specialise (Human development/ human needs?

- PF2 - upregulation/ down regulation of enzymes (examples of mechanism stated?)

- PF3 - temporal expression e.g. according to human developmental phase (fetal haemoglobin)/ respond to environment changes (growth factors/ cell division/apoptosis)

- PF4 - spatial expression (different cell types requires different proteins to form different structure to perform different functions)

- PF5 - Why there is protein variety, classification of proteins/different class of protein for different functions/purpose (Examples: Transport/ enzymatic activity/ cell-to-cell recognition/ receptor binding/ cell signalling/intercellular joining/ stabilizing proteins/ respiration)
- PF6 - environmental factors (e.g. exposure to sun, increase melanin production)
- PF7 - post translational modification → alternative splicing → different mRNA (also nucleic acid) formed (support variety of proteome)
- PF8 - post translational modification → chemical modification such as phosphorylation/ glycosylation/ addition of lipids

(discuss against human's proteome is constantly changing)
- PA1 - the need for constancy in the proteome for the proteins involved in fundamental life processes, such as respiration idea of conserved proteins/ housekeeping proteins (no matter the time/ space in a human)

- PA2 – due to degeneracy of genetic code, there might be mutations in the genome but there will be no change in the mRNA codon→ amino caid especially when the mutation occurs on the wobble base of the DNA base triplets

QWC (address all sections)

Examiners' comments
Those candidates who appreciated the meaning of 'proteome' were able to engage effectively with this title and make links to a range of topic areas with which they were familiar. Stronger candidates used a clear line of explanation and argumentation to form the bulk of the essay content.

Relevant issues for consideration included the significance of mutation, the generation of variation during meiosis, various forms of genetic modification and a discussion of the possible consequences of these changes to the genome. For the proteome, candidates could have discussed the need for changing protein expression as cells grow and specialise. A small number of candidates discussed the need for constancy in the proteome for the proteins involved in fundamental life processes, such as respiration. A discussion of conserved proteins would have improved the quality of responses of some candidates.

A few candidates misread the question and answered it in terms of 'the human genome and proteome' instead of 'a human's genome and proteome'. Although many relevant issues were still explored, it meant that the essay had an emphasis on evolutionary change in a population as opposed to the uniqueness of the individual (examiner's interpretation).

Reference to **transport**
Hydrophilic channels involved in facilitated diffusion;
For specific ions and polar molecules to pass through;
Channel proteins have fixed shape/ carrier proteins can undergo rapid changes
In shape in oprder to transport specific sunstances;
Na^+ and K^+ pumps involved in active transport;
Actively pumps $2K^+$ into cell and $3Na^+$ out of cell against concentration gradient;

Reference to **<u>enzymatic activity</u>**
Enzyme embedded in membrane with its active site exposed to substances;
E.g. inner membrane of mitochondria contains ATP synthase;
For the synthesis of ATP/ in oxidative phosphorylation;

Reference to **<u>cell to cell recognition</u>**
Glycoproteins serve as identification tags;
Antigens serve as markers;
To recognize specific cells fo development of tissues and organs;
Reference to antibody-antigen binding
Antigens act as cell identity markers
Foreign antigens can be recognized and attacked by immune system;

Reference to **<u>receptor binding</u>**
Chemically signally between cells;
Protein receptors that have binding site with specific shape;
Binds to a transmitter (hormone or neurotransmitter);
Causes conformational change to the protein;
That relays message to the inside of the cell
E.g. receptor on post-synaptic membrane/ receptors for cyclic AMP/ receptors for signal transduction pathway/ receptors for hormonal regulation (insulin receptors);
protein phosphorylation cascade in cell signalling

Reference to **<u>intercellular joining</u>**
Proteins in adjacent cells are hooked together in various kinds of junctions;
Tight junction: membrane fused to prevent leakage of fluid;
Anchoring junction: fasten cells together into strong sheets;
Gap junctions: provide channels between cells for passage of ions and small molecules;

Reference **<u>to stabilizing proteins</u>**
Carbohydrates chains of glycoproteins and glycolipids projects into aqueous environment surrounding cell;
Form hydrogen bonds with water molecules which stabilize the membrane;

[Total:25]

5 **(a)** Explain why more energy is released from the oxidation of triglycerides as compared to the same unit mass of glucose and outline the problems of using triglycerides as the main respiratory substrate instead of glucose in animals. [10]

- triglycerides contain **two times more hydrogen atoms** / **greater ratio of hydrogen to oxygen atoms** / **more C-H bonds** as compared to the same mass of glucose;
- Can be used to generate **greater** number of **NADH** and **FADH** during **Kreb's cycle** / **dehydrogenation reactions**;
- Release **more** high energy **electrons** to the electron transport chain;
- Generate more ATP by **oxidative phosphorylation**;

- Glucose is **polar / hydrophilic** due to presence of many hydroxyl groups;
- Glucose is therefore soluble in aqueous medium, can easily be transported in blood and cellular cytoplasm;

- Small molecule so can diffuse through blood vessels / across the cell surface membrane through the glucose carrier proteins / via facilitated diffusion;

- Triglycerides are **non-polar / hydrophobic** due to the fatty chains / hydrocarbon chains;
- Triglycerides are insoluble in aqueous medium, cannot be easily transported in aqueous medium / needs to form lipoprotein complexes with proteins to be transported in blood;
- Cannot easily cross the cell surface membrane / needs to enter the cell via **receptor-mediated endocytosis**;

- AVP;

- QWC (one point from each section);

(b) Discuss the inter-relationships of membranous organelles in eukaryotic cells and outline the implication of absence of membranous organelles in prokaryotic cells. [15]

Protein synthesis and secretion

- **Nucleus** : site of **transcription** (of DNA to form RNA)

- Using **RNA nucleotides/ RNA polymerase** and **DNA template** found in the nucleus

- RNA (accept merely mention mRNA?) formed leave via **nuclear pores** to enter **cytoplasm**

Ignore reference to free ribosomes

- **Ribosomes** found on **rough endoplasmic reticulum** carry out **translation** of proteins of the endomembrane system OR proteins to be secreted from the cell

- Route from protein: Proteins are then packaged into **transport vesicles** → **Golgi body**

- Within the Golgi body, proteins from RER are further **modified**, **sorted** and **packaged**

- **Lipids** from **smooth ER** are also **modified**, **sorted** and **packaged**

- From Golgi body to **cell surface membrane** via **secretory vesicles** / to other parts of the cell via the **Golgi vesicles**

- *Award once only* for **pinching off/ budding** of vesicles from organelles + **fusion** of membrane of vesicles with the destination organelle

  E.g. Membrane of transport vesicles fuse with the membrane (of the *cis* face) of the Golgi body and deposit their proteins within the lumen of the Golgi body OR secretory vesicles which bud/ pinch off from (the *trans* face of the) Golgi body to fuse with CSM

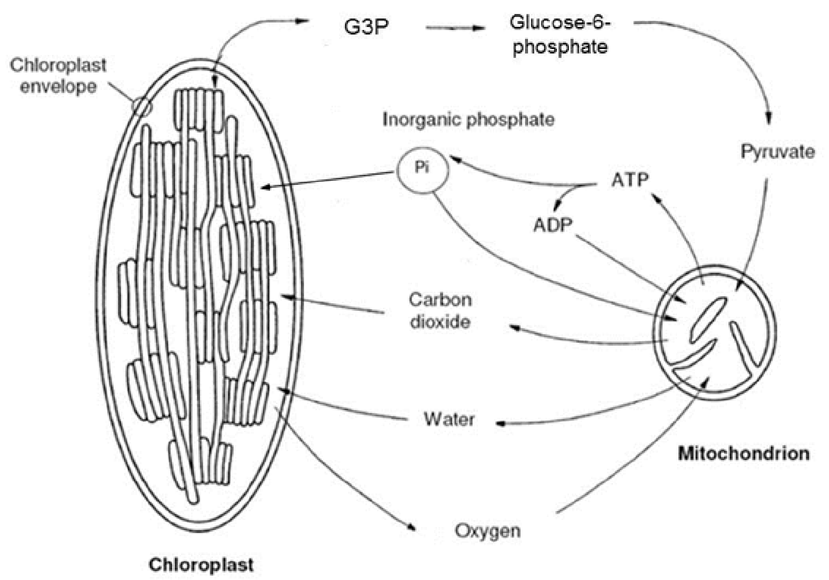
- *award once only for movement of vesicles :* With the aid of **microtubules,** the secretory vesicles will move to the cell surface membrane

Mitochondria / chloroplast with RER (TYS)

- mitochondrion **produces ATP** for rough endoplasmic reticulum for protein synthesis/translation

- RER synthesises proteins/ enzymes which are required for respiratory reactions in mitochondria/ photosynthesis in chloroplast

Mitochondria and chloroplast (2013 Promos) in plant cells



- **G3P** formed in **chloroplast** due to **light-independent reaction/ Calvin cycle** (converted to glucose)
- **pyruvate** (accept glucose?) **undergo aerobic respiration** / the **link reaction, Krebs cycle and oxidative phosphorylation** to **be broken down for ATP production in mitochondria**
- Oxygen is formed (In thylakoid space) as a **product of** the **photolysis** of **water** during **light-dependent reaction**
- Oxygen (diffuses out of the chloroplast and into the mitochondrial matrix) acts as the **final electron acceptor** (in the electron transport chain to produce water) in oxidative phosphorylation
- Carbon dioxide **produced** in aerobic respiration during **Krebs cycle** and **link reaction** / by oxidative decarboxylation
- (Diffuses into the stroma of the chloroplast) for **carbon fixation** in the **light independent reaction/Calvin cycle**;

Lysosomes

- Vesicles containing **digestive/ hydrolytic enzymes**
- Which **bud off / pinch off** from the **Golgi apparatus**
- **Phagocytosis**: Substances taken in by the cell in a vacuole/ **phagocytic vesicles** are fused with lysosomes
- **Autophagy**: A damaged organelles or small amount of cytosol becomes surrounded by a membrane, which then fuses with lysosome to form a vacuole in which the unwanted materials are digested.

Implication of absence of membranous organelles in prokaryotes

- DNA **not enclosed** by **nuclear membrane**
- This allows **transcription** and **translatio**n to take place **simultaneously**
- No RNA processing (since transcription and translation occurs

simultaneously)

→ no **post-transcriptional control** of gene expression

- **Fewer genes / smaller genome**: since there is no need to encode transport proteins/ signaling proteins/ etc on membranes of organelles
- Protein secretion (no exocytosis/ endocytosis) can only occur through **channel proteins / facilitated diffusion**
- Stalked particles / respiration / photosynthesis (for cyanobacteria) occurs only on **cell surface membrane**

[Total: 25]